

Reduction in Perceived Stress in Healthy Women Older Than 30 Years Following a 24-Week Resistance Training Program: A Pilot Study

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ABSTRACT

Background: Resistance exercise can help counteract estrogen-dependent changes in the musculoskeletal systems of older women, but its impact on female psychosocial health has not been well characterized.

Objective: Complete pilot study measuring psychosocial variables in healthy women older than 30 years following a weight-lifting program.

Study Design: Nonrandomized study of psychosocial measures in 13 female experimental participants (mean = 45.5 ± 8.4 years) with control group comparison (n = 5; 52 ± 11.8 years).

Methods and Measures: Experimental participants without prior lifting experience completed a resistance exercise program for 24 weeks. Psychosocial and physical variables were measured before and after the program. Psychosocial surveys measured self-esteem, perceived stress, mastery, and body satisfaction. Physical variables included body mass index and muscle strength using a 1-repetition maximum test. Control participants completed identical surveys but no resistance exercise. Outcome variables were analyzed with paired *t* tests.

Results: Perceived stress significantly decreased in experimental participants following the program ($P < .05$). Other psychosocial variables showed favorable trends for women compliant with the program, but the difference in pre- and postmeans were not significant. No psychosocial variables changed significantly for controls. There was a significant increase in body mass index and the 1-repetition maximum test for the leg press and lat pull-down machines ($P < .05$).

Conclusions: This pilot study demonstrated a significant decrease in perceived stress in a group of women older than 30 years completing a 24-week resistance exercise program. Research examining psychological benefits of

exercise is extremely important to help promote healthy lifestyle choices and preventative guidelines.

Key Words: older women, psychosocial health, weight-lifting

INTRODUCTION

Stress has profound effects on physical and psychosocial health. Research has shown that chronic psychological stress and/or elevated levels of glucocorticoids have negative consequences on a wide range of physiological systems including hippocampal volume, cognitive performance, insomnia, and immune function.¹⁻³ The impact of chronic stress is becoming better understood at the molecular level: Epel et al⁴ showed in a group of healthy premenopausal women that the duration of perceived stress was significantly correlated with markers of cellular aging. The authors examined the length of telomeres, which are repeated DNA sequences located at the end of chromosomes, which shorten as cells divide and age. Specifically, Epel et al found that women who were caregivers for chronically ill children demonstrated decreased telomere length and higher levels of oxidative stress markers.⁴ As the impact of psychological stress on the human body is better characterized, it is crucial to explore how preventative interventions such as exercise can have positive consequences for chronic stress and women's health.

One significant source of stress for some individuals involves age-related changes in the body and potential loss of independence. As men and women age, they experience a decrease in skeletal muscle mass or sarcopenia.^{5,6} Age-dependent loss of skeletal muscle involves a variety of mechanisms including altered protein synthesis and changes in levels of hormones and inflammatory chemicals.⁷ Decreased muscle strength is associated with impairments in gait, balance, and activities of daily living,^{8,9} and severe sarcopenia has been shown to be a risk factor for physical disability in older women.¹⁰ This

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cycle of decreased strength and function not only has obvious physical consequences, but can also affect psychosocial health and overall quality of life.

On the basis of personal observation and clinical experience, women tend to participate in cardiovascular exercise options more often than resistance training programs. According to a 1998 article by Ebben and Jensen,¹¹ a muscular physique typically has been associated with masculinity and a thin, small build has been considered feminine, which has helped deter women from any kind of muscle-building activities. These authors discuss variables that tend to decrease effective strength training in women: misconceptions about weight gain and muscle hypertrophy, fear of injury, and hesitancy to train with high load or intensity. Resistance exercise has been shown to cause significant increases in strength and skeletal muscle hypertrophy in older men and women.^{12,13} In one study by Melov et al,¹⁴ the completion of a 26-week resistance exercise program not only significantly increased strength in the older subjects but also changed the pattern of gene expression in the mitochondria of skeletal muscle cells. Specifically, the authors found that the expected shift to a higher expression of “aging” mitochondrial genes in older muscle was partly reversed by the exercise program and a younger pattern of gene transcription was found in the mitochondria following the program.¹⁴

It is an important goal of physical therapists to promote resistance training in women, especially to help maintain bone health in older women. Health care costs to manage osteoporotic fractures are in the billions, and the annual cost is projected to be \$25.3 billion by the year 2025 to manage more than 3 million fractures.¹⁵ According to a 2002 report from the National Osteoporosis Foundation,¹⁶ 30 million women older than 50 years were categorized with low bone mass or osteoporosis. This report predicts that the number of women will increase to 41 million by the year 2020. Several studies have shown that exercise does, in fact, improve bone health and other important physical outcomes.^{17,18} Nelson et al¹⁷ demonstrated that a 12-month program of high-intensity strength training in postmenopausal women maintained total body bone mineral content and improved total muscle mass, strength, and dynamic balance. A meta-analysis by Wolff et al¹⁸ examined the effect of various exercise programs on bone mass in pre- and postmenopausal women and concluded that exercise can prevent or reduce approximately 1% bone loss in the lumbar spine and femoral neck. Exercise interventions that improve bone health may also successfully impact psychosocial outcomes: research has shown that health-related quality of life is significantly correlated with the number and type of osteoporotic fractures in postmenopausal women.^{19,20}

There has been substantial research demonstrating psychosocial improvement in clinical and nonclinical populations following aerobic exercise programs.²¹⁻²³ Weight training programs have also been correlated with various psychosocial improvements.^{24,25} Tsutsumi et al²⁴ found that 12 weeks of resistance training in healthy older adults (mean age of 68 years) had psychological benefits. Specifically, both the high-intensity and low-intensity strength training groups improved in positive and negative mood and physical self-efficacy. In another study examining male law enforcement personnel completing a 16-week weight training program, there were significant improvements in anxiety, depression, hostility, and somatization in the participants.²⁵

Overall, the research examining the psychosocial impact of resistance training in healthy older women is limited. The purpose of this pilot study was to measure changes in psychosocial outcomes (self-esteem, perceived stress, mastery, and body satisfaction) and basic physical variables (strength and body mass index [BMI]) in a group of women older than 30 years following a 24-week training program. Based on the previous research demonstrating the ability of aerobic and resistance training exercise to improve skeletal muscle strength¹²⁻¹⁴ and psychosocial outcomes, including anxiety, depression, and the ability to cope with stress,²¹⁻²⁵ our hypothesis was that women who were compliant with the resistance exercise program would demonstrate significant improvements in both physical and psychosocial variables compared with women who were not compliant with the resistance exercise program. Specifically, we predicted that women who completed at least 70% of the exercise program sessions would demonstrate increased physical strength in major muscle groups. In addition, we expected that these women would report significantly lower perceived stress as a result of their participation in the exercise sessions. Finally, we predicted that women who completed the exercise program would report higher self-esteem, improved body satisfaction, and increased sense of mastery compared with women who did not participate in any resistance exercise.

METHODS AND MEASUREMENTS

Participants

Of the possible 228 women eligible to participate in the study, a sample of 13 female faculty and staff members at a small, private liberal arts university in the southern United States participated in this pilot study. Criteria for eligibility included being older than 30 years, not pregnant, no prior experience in weight training, and no significant medical history or diagnosis that would limit participation in the

program. All participants were required to obtain medical clearance from a registered nurse or primary care physician to begin the study. A control group of 7 women meeting the same exclusion criteria as the experimental group was also recruited through the same means. Control subjects were asked to refrain from doing any resistance training exercise for the duration of the study. Before the initiation of any study procedures, all subjects completed the informed consent and signed a debriefing statement explaining the purpose of the study and contact information for any future questions or discussion of the study's results. The institutional review board at Queens University of Charlotte approved this nonrandomized control study.

Participants in the experimental group were given paperwork, which included log sheets to record the frequency and intensity of exercise, exercise tips and guidelines (Appendix 1), an exercise instruction guide with photographs demonstrating exercises (Appendix 2), and a questionnaire to collect basic information on exercise, sleep, and dietary habits. At this initial meeting, participants were assigned numbers to maintain anonymity throughout the duration of the study. A licensed physical therapist explained and demonstrated all exercises provided in the guide and subsequently checked participants for proper form on all exercises. Important precautionary guidelines were also discussed, and all questions were answered. Each subject completed the exercise sessions independently in the venue of their choice.

The exercise program consisted of 9 exercises and was designed to effectively target the major muscle groups as follows: shoulder (Arnold Press), triceps (triceps kickback), biceps (bicep curl), back (dumbbell row and lat pull-down), chest (chest press), leg (squat and lunge), and abdominals (crunch or standing stabilization variation). Participants were asked to complete each of the 9 exercises with 3 sets of 8 to 10 repetitions for each exercise. They were instructed to complete at least 2 weight training sessions per week of approximately 30 to 40 minutes in length and rest a minimum of 2 days between lifting sessions. Participants were instructed to lift a weight that felt challenging by the sixth or seventh repetition and to gradually increase their weight when the 10 repetitions became too easy. Strength training protocols in the literature vary with the number of sets and repetitions for each exercise and duration of the total program.^{12-14,17,18,24} The protocol used in Melov et al¹⁴ was chosen for this study because the 2 sessions per week of 3 sets of 10 repetitions produced significant strength increases in the subjects after 26 weeks; moreover, this lifting frequency would be expected to produce higher compliance as compared with a program requiring 3 sessions per week. Although

the physical therapist was available for questions at any time during the program to guide participants as needed, no participants contacted the physical therapist to be present at any exercise sessions. The length of the exercise program was 24 weeks, and a minimum of 34 sessions (70%) was considered compliant with the program.

During the week before the beginning of the exercise program, the experimental and control groups completed an online survey with the psychosocial scales that coincided with the initiation of the exercise program for the experimental group. Surveys were completed again by both groups at 6 months after the completion of the initial survey. Lastly, all participants completed a follow-up questionnaire approximately 3 months after the end of the exercise program and completion of second psychosocial survey. Start dates for the exercise program were recorded for all participants in the experimental group to coordinate the initial (preexercise) and final (postexercise) measurements.

Psychosocial Measurements

Psychosocial scales and follow-up questions were administered to both groups by using an online Survey Monkey, which was password protected. The pre- and postexercise surveys consisted of established psychosocial scales measuring self-esteem, perceived stress, mastery, and body satisfaction. The postexercise survey also asked about the number of positive and negative life events that participants had experienced during the previous 6 months. Experimental participants were also asked about how their clothes fit after weight training, whether they exercised individually or in groups, and whether or not they enjoyed the program.

Self-esteem

The Rosenberg Self-Esteem Scale was used to measure self-esteem and consists of 10 items.²⁶ Research has demonstrated consistent reliability and validity of this scale as a measure of self-esteem in various populations.^{26,27} Participants indicated the level to which they agree or disagree with each of 10 statements regarding self-esteem. Total scores were obtained by reversing individual scores on items 1, 3, 4, 7, and 10 and then summing across all items. A low score indicates low self-esteem and the maximum score is 40, corresponding to a high level of self-esteem. For this sample, the Cronbach alphas for pre- and postmeasures were acceptable (0.93 and 0.90, respectively).

Perceived Stress

Perceived stress was measured using the Perceived Stress Scale.²⁸ Research has demonstrated adequate reliability and validity of this scale as a measure of

perceived stress.²⁸ The Perceived Stress Scale consists of 14 items, and total scores are obtained by reversing individual scores on the 7 positive items and then summing across all items. This scale measures the degree to which situations in one's life are considered stressful. A high score indicates high perceived stress with a maximum score of 56. The Cronbach alpha for the pretest measure was 0.93 and for the posttest measure it was 0.96.

Mastery

Mastery was measured using the Pearlin-Schooler Mastery Scale, which consists of 7 statements regarding perceptions of mastery.²⁹ Research has demonstrated consistent reliability and validity of this scale as a measure of mastery in various populations.²⁹⁻³¹ Total scores were determined by reversing individual scores for items 1 through 5 and then summing across all items. Low scores indicate low mastery and the maximum score of 28 represents a high level of mastery or greater perceived self-control. For this sample, the Cronbach alphas for the pre- and posttest measures were 0.91 and 0.90, respectively.

Body Satisfaction

The Body Satisfaction Scale was used to measure perceptions of body satisfaction and dissatisfaction.³² Research has demonstrated adequate reliability and validity of this scale as a measure of body satisfaction.^{32,33} The Body Satisfaction Scale evaluates 16 body parts, with 8 items involving the head and 8 involving body parts below the head. Participants rated their level of satisfaction/dissatisfaction with each body part on a 7-point Likert scale. Lower numbers indicate satisfaction and higher numbers indicate dissatisfaction. Total scores are determined by reversing individual scores on all items and then summing across all items. The maximum total score is 112, which represents the highest level of body dissatisfaction. The Cronbach alphas were acceptable for both the pre- and posttest measures (0.82 and 0.93, respectively).

Physical Measurements

Height and weight were measured for all experimental subjects to calculate the BMI using the equation $BMI = \text{weight (lb)} / [\text{height (in)}]^2 \times 703$.³⁴ Strength was measured for 3 different muscle groups using a 1-repetition maximum strength test (1-RM): quadriceps/gluteus muscles, biceps, and latissimus dorsi. The 1-RM is defined as the maximum weight that can be lifted for one repetition with correct form and is one method used in the literature to measure muscle strength.³⁵ Lower extremity strength was tested on a seated horizontal leg press machine, biceps strength on a seated biceps curl machine, and back strength

on a seated lat pull-down machine. This value was determined for most participants within 5 repetitions of incremental weight increase and a short rest period of approximately 30 seconds was given between each repetition until the 1-RM was reached. All physical measurements were taken prior to the program (pre-exercise) and once following the 24-week exercise program (postexercise). Exercise log sheets were collected at the postexercise meeting with participants.

Follow-up Survey

All participants filled out a follow-up questionnaire 3 months after the conclusion of the exercise program. It asked for basic demographic information and feedback on the program. The experimental group indicated if they were still strength training 3 months after the program; for those who were no longer strength training, they were asked why they had stopped participating in the program. Furthermore, they were asked to comment on any positive or negative changes they experienced during the resistance exercise study.

Statistical Analysis

All data were analyzed using the Statistical Package for the Social Sciences (version 17, SPSS, Chicago, IL). For the experimental group, outcome variables were analyzed for only those participants who completed at least 70% of the total 48 sessions. For the control group, data were analyzed for only those participants who completed both pre- and postsurveys. Because of the small sample size of the experimental and control groups in this pilot study, within-group analyses using paired *t* tests were performed to examine whether the resistance exercise program significantly changed physical and psychosocial variables. The level of significance for all analyses was set at $\alpha = .05$.

RESULTS

Descriptive statistics for the experimental and control groups are shown in Table 1. Overall, the majority of women were white and married, with a graduate degree and earning an income more than \$50 000, and all participants were nonsmokers. The initial control group was reduced to 5 because of missing data from poor compliance with survey completion and 1 of these control subjects failed to provide data for the descriptive statistics. The 9 participants in the experimental group who completed a minimum of 70% of the total exercise sessions were analyzed for psychosocial and physical variables. The other 4 experimental participants did not meet this level of program compliance because of various medical reasons unrelated to the exercise program.

Table 1. Descriptive Statistics of Participants

Variable	Experimental (n = 13)	Control (n = 4)
Age		
Mean (SD), y	45.5 (8.37)	52 (11.75)
Range, y	33-59	37-64
Race		
White	12 (92.3%)	4 (100%)
Black	1 (7.7%)	0 (0%)
Marital status		
Married	8 (61.5%)	3 (0%)
Single	4 (30.8%)	0 (0%)
Separated	1 (7.7%)	0 (0%)
Widowed	0 (0%)	1 (25%)
Education		
BA/BS	1 (7.7%)	0 (0%)
Graduate degree	12 (92.3%)	4 (100%)
Income		
\$20 000-\$50 000	4 (30.8%)	2 (50%)
>\$50 000	9 (69.2%)	2 (50%)
Abbreviations: BA, bachelor's of arts degree; BS, bachelor's of science degree.		

Psychosocial Variables

The mean scores for each psychosocial scale changed in the expected direction for the experimental participants successfully completing the resistance exercise program (Table 2). Scores for the self-esteem, mastery, and body satisfaction variables all improved, but not significantly. Perceived stress, however, significantly decreased in the experimental subjects with mean scale scores decreasing by 6.9 points from 22.3 to 15.4 ($P < .05$). None of the psychosocial variables changed significantly for the control group (Table 3).

Physical Variables

Following the 24-week program, BMI increased from 24.8 ± 2.9 to 25.4 ± 3.2 (mean $\pm$ SD) in

Table 2. Measurements of Psychosocial Variables for Experimental Group (N = 9)

Variable ^a	Preexercise Mean (SD)	Postexercise Mean (SD)	<i>P</i> ^b
Self-esteem	32.8 (4.8)	34.3 (4.6)	.222
Perceived stress	22.3 (7.4)	15.4 (8.9)	.023
Mastery	22.4 (3.4)	24.1 (2.9)	.165
Body satisfaction	89.8 (13.7)	96.3 (13.5)	.158
^a Measures included Rosenberg Self-Esteem Scale, Perceived Stress Scale, Pearlin-Schooler Mastery Scale, and Body Satisfaction Scale. Nine experimental participants who had completed at least 70% of total sessions were included in analysis.			
^b <i>P</i> value for paired <i>t</i> test of means.			

Table 3. Measurements of Psychosocial Variables for Control Group (N = 5)

Variable ^a	Baseline Mean (SD)	Post-6-Month Mean (SD)	<i>P</i> ^b
Self-esteem	36.2 (3.8)	34.4 (6.6)	.305
Perceived stress	14.2 (9.8)	17.2 (8.9)	.284
Mastery	25.2 (3.1)	24.0 (4.5)	.222
Body satisfaction	87.4 (7.02)	92.2 (16.6)	.492
^a Measures included Rosenberg Self-Esteem Scale, Perceived Stress Scale, Pearlin-Schooler Mastery Scale, and Body Satisfaction Scale. Five control participants were included in analysis.			
^b <i>P</i> value for paired <i>t</i> test of means.			

the 9 experimental participants, which approached significance ($P = .088$). For the 1-RM measurements, there were significant improvements in lower extremity and back strength (Table 4; $P < .05$). On average, the experimental group was able to lift 58 pounds more on the leg press and 9.3 pounds more on the lat pull-down machines after the program. Measurements of 1-RM for the biceps curl machine indicated no significant increase in biceps strength (Table 4; $P = .462$).

Subjective Comments

Overall, the subjective feedback from the exercise group participants regarding the resistance training program was positive. Of the 11 persons in the exercise group who completed the follow-up survey, 8 were still participating in their exercise program. All but 1 of the respondents replied that they had enjoyed the program; the remaining participant answered that she had enjoyed it only "sometimes." Approximately half of the 11 respondents said that their clothes fit more loosely and the other half said that their clothes fit the same. One participant shared that her total cholesterol level had dropped substantially and another subject reported self-terminating her antidepressant medication. Multiple e-mails from the latter participant described her strong, newfound enthusiasm about lifting weights.

Table 4. Measurements of Physical Variables for Experimental Group (N = 9)

Variable ^a	Preexercise Mean (SD)	Postexercise Mean (SD)	<i>P</i> ^b
Lower extremity strength	187.6 (35.4)	245.6 (48.4)	.003
Biceps strength	22.6 (7.4)	24.4 (5.6)	.462
Back strength	76.3 (6.9)	85.6 (5.0)	.001
^a Measurements given in mean pounds lifted for 1-repetition maximum strength test on seated horizontal leg-press, biceps curl, and lat pull-down machines. Nine experimental participants who had completed at least 70% of total sessions were included in analysis.			
^b <i>P</i> value for paired <i>t</i> test of means.			

COMMENT

In this pilot study examining psychosocial and physical outcomes of a resistance exercise program in older women, there was a significant improvement in psychosocial and physical variables. The main psychosocial result was a significant decrease in perceived stress in the experimental participants who completed at least 70% of the exercise program. The other variables of mastery, self-esteem, and body satisfaction showed trends in the favorable direction, although they were not statistically significant.

The decrease in perceived stress following the resistance training program is a very important finding, especially for women's health, as this type of exercise offers important physiological benefits for older women.^{13,17,18} Aerobic exercise programs have been related to positive changes in psychosocial variables such as stress and anxiety in other populations.²¹⁻²³ Chronic stress can negatively impact physiological systems through a variety of mechanisms, including catecholamines, corticosteroids, and inflammatory chemicals. At the genomic level, chronic stress has been linked to accelerated aging and increased loss of telomeres.⁴ Exercise has been shown to impact levels of neurochemicals²³ and correlated with decreased markers of chronic inflammation in populations of postmenopausal women.^{36,37} In our study, 2 participants believed that their reduced level of cholesterol and cessation of antidepressant medication were related to completion of the resistance exercise program. A better understanding of the relationships underlying chronic stress, exercise, and health outcomes is crucial for future research to optimize physical therapy interventions and the ability to help prevent or reverse disease processes.

In addition to perceived stress, an individual's perceptions of control and self-efficacy are important psychosocial factors to consider for health outcomes. Although this study showed an increase in the mean postexercise mastery scores for the experimental group, this change was not significant. Levels of mastery or perceived control have been shown to be related to quality of life and self-ratings of health in individuals with a chronic illness or disability.^{38,39} Moreover, research examining educational interventions designed to improve perceptions of control and self-efficacy in patients has demonstrated positive impacts on psychosocial variables and other health outcomes.⁴⁰ However, perceived control has a complicated relationship with health status and this psychosocial variable interacts with many other factors to impact behavior and health outcomes.⁴¹ It is important to explore how consistent exercise programs may improve a woman's sense of empowerment and perceived control over her life and health status.

One component of exercise that may play a role in any potential psychological benefits is exercising with other people versus individually. Hughes et al⁴² reported that a 12-week individual exercise program did not have a major positive impact on psychological variables in their study of 14 sedentary men. One of the factors suggested by the authors that may be important for psychological benefit is socialization during exercise. In another study that reported psychological improvement in adults completing a 12-week aerobic exercise program, the authors discuss that "social support" may have played a role in the psychosocial improvement of the exercise subjects versus control subjects.²¹ Some subjects in our study commented that they enjoyed working out with others versus individually and would have preferred a group exercise option. When designing future studies, the socialization variable is an interesting factor to consider.

Although the focus of this pilot study was on psychosocial outcomes, we also completed basic measurements of physical variables in the experimental group. Although the mean BMI increased after the program, this change was not significant. Subjectively, half of the exercise group reported that their clothes fit more loosely, which can be a more appropriate feedback during an exercise program versus scale measurements of weight. Results showed a significant improvement in strength for 2 of the 3 muscle groups measured. Specifically, the postexercise 1-RM was significantly higher for the lateral pull-down and leg press machines but not the biceps curl. A number of the participants found the biceps curl machine uncomfortable, which may have affected performance on the machine. It is worth noting that an overall increase in confidence and comfort level with resistance training may have played a role in the strength gains measured. For all compliant participants, the amount of weight lifted increased for the various exercises over the 6-month duration of the program. Future studies would benefit from a more detailed lifting protocol for participants based on percentages of 1-RM with appropriate progression of resistance and from incorporating more valid and reliable measurements of muscle strength, such as a dynamometer. Racial diversity of the sample is another variable to consider as some research discusses that the age and race of a woman should be considered when designing strengthening exercise programs.⁴³

Other important factors that may influence interpretation of this study are the lack of randomization, initial psychosocial baseline of the participants, and the potential impact of outside life events on psychosocial variables. Our exercise group was made up of volunteers, and the lack of random allocation to groups is a major limitation of this study. In their

meta-analysis of randomized and nonrandomized control trials, Wolff et al¹⁸ found higher treatment effects in nonrandomized control trials suggesting that the lack of randomization can be a strong confound. Selection bias can be a confounding variable as people who volunteer for exercise studies may be more motivated and likely to demonstrate positive psychosocial effects.⁴² For our experimental group, comparisons of the mean preexercise scale scores to the upper/lower limits of each psychosocial scale indicated that our sample did not present with especially low levels of mastery and self-esteem or high levels of perceived stress at baseline. Norvell and Belles²⁵ found that in comparison with the participants who finished a resistance exercise program, the subjects who dropped out had significantly greater levels of anxiety, depression, and hostility at baseline. Because our sample appeared to have relatively strong psychosocial health at baseline, the results of this study may not be indicative of realistic results for a larger population. Another important variable to consider is the impact of other life events on perceived stress and other psychosocial variables. Both groups were evaluated over the same 24-week interval to control for any global or local events that might substantially impact the psychosocial variables in participants. It can be difficult to interpret psychosocial outcomes of exercise programs when other factors play an important role in psychosocial health.

There are several limitations in this study that will help with future research design. There was limited statistical power due to the small sample size. We found it difficult to attract a large sample size and groups were self-selected on the basis of the fact that women were motivated to try resistance exercise compared with women who were less motivated. As discussed previously, randomized allocation of volunteer subjects will limit selection bias and improve future research design. Some subjects had poor compliance with filling out the log sheets in detail, which limited our ability to analyze progression of workout intensity. The age range of our experimental group was 33 to 59 years, which encompasses a wide spectrum of age-related changes in skeletal muscle and hormone levels. Future studies could be performed on older women of more similar age or only postmenopausal participants to help decrease physiological variability. For those who did participate in our experimental group, scheduling was often an issue and individual versus group exercise sessions with a more focused progression of resistance was not feasible in this pilot study.

CONCLUSIONS

Research has demonstrated that the benefits of exercise are numerous, both physical and psychosocial. In

this pilot study, women with no prior lifting experience showed significant decreases in perceived stress and increases in muscle strength following a 24-week resistance exercise program. Although the control group was small in this study, we believe that the lack of significant changes in the pre- and postscales for our control group combined with the subjective comments suggests that the resistance exercise program did play a role in the significantly reduced levels of perceived stress reported by our experimental participants. For future research, our goal is to explore psychosocial outcomes in combination with physiological markers of oxidative stress and inflammation in older women completing resistance exercise. Examining how exercise interventions may influence the relationship between perceptions of stress, control and self-esteem, and cellular markers of stress is one approach to better understand the powerful impact of positive health choices.

Clearly, stress management is of extreme importance because of our fast-paced society and increasing evidence of the role chronic stress plays in the pathogenesis of various diseases. Cost-effective preventative measures such as exercise programs are more important than ever as our indebted government faces ever-rising health care costs. With the aging baby boomer population, an increasing number of individuals, especially women, will be at risk for osteoporosis and associated fractures. Further research should examine the implications of weight training and other types of exercise for stress reduction and disease prevention in women. It is crucial for clinicians to promote preventative exercise programs and reinforce lifestyle choices that maintain both physical and psychosocial health and optimize independence and quality of life. ○

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APPENDIX 1. Exercise Tips and Guidelines

Here are some things to keep in mind that will help you make the most of a strength training program:

1. *Warm up for at least 5 minutes before you begin lifting.* Think of your muscles as rubber bands. What would happen if you stretched a frozen rubber band? It would break! You can do anything you prefer, as long as it gets your heart rate up and your muscles warm. Something as simple as walking briskly to the fitness center would be acceptable.
2. *Challenge yourself.* As you get used to lifting, gradually increase the amount of weight you use every 2 to 4 sessions. This will help you see better results and become stronger. Use an amount of weight that will cause your muscles to burn slightly by the end of the set. If you do not challenge the muscles, they will not be stimulated to grow new fibers and get stronger. At the end of the 2 sets for a particular exercise, you should feel like you would have some difficulty doing it again.
3. *Muscle soreness.* With strength training, it is normal to feel muscle fatigue and soreness. If you have done a good workout, your arms and legs will feel a bit weak after the exercises because you have stressed them. Over the following 1 to 2 days, it is normal to feel muscle soreness, which is not a sharp pain, but a low-level soreness as the muscles rebuild from the small tears caused by the lifting exercises. It is similar to the soreness you experience after a new and prolonged activity, such as moving many boxes or painting a large room. The soreness will go away as you become accustomed to the specific weight that you are lifting. You will then need to increase the weight level to continue getting more benefit from the workout. This will result in stronger muscle that burns more calories at rest—this is part of the increased metabolism benefit of weight training. You should *not* experience any sharp pain in any joint or muscle during the exercises or following the workout. Sharp pain may indicate improper lifting form or an overuse injury from the new exercises. Contact us to discuss any questions or concerns about how your body feels after you start the program.
4. *Always keep your form in mind when you are lifting.* Proper form is extremely important in avoiding injury. Dr Koplas and I will demonstrate how to properly perform each exercise and you will have a packet with explanations and pictures, as well. If you forget a particular exercise and we are unavailable, you will also be able to refer to a list of links for online videos to view the proper form.
5. *Remember to eat.* When you lift weights, your muscles are essentially torn and need to rebuild. Try to have at least a small snack within an hour before or after your workout. Whatever you eat should include some protein, carbohydrates, and a small amount of good fat. Protein is extremely important in building muscle, carbohydrates will give you energy, and a small amount of healthy fat is an essential part of the human diet. Yogurt, granola bars, trail mix, nuts, and fruit are some good foods to consider.
6. *Stretch out.* Whenever you have a chance (on the days you do not lift, maybe), do some simple stretches. This will help your muscles heal. It will also help you get rid of any soreness that you may experience. Stretching is also a great way to avoid injury.
7. *Do not work out on consecutive days.* Your muscles need time to rest and recover from one session before you exert them again. Ideally, you should take 2 days off between strength training sessions if you are sticking to 2 exercise sessions per week.
8. *Track your progress.* You will have a personal log with sheets for each time you work out. It is important to record the date, the weight you are using, and the number of sets and reps. Please also record, in as much detail as possible, any other additional exercise you do. Keeping a log will let allow you to quantify your results, see your achievements, and stay motivated! It will also help Dr Koplas and me to see how each of you is progressing with the program and decide whether or not adjustments are needed.
9. *Avoid the scale.* Weighing yourself is not the best way to track your progress during a strength training program. We recommend choosing a tight-fitting outfit and noting how it fits every 2 to 3 weeks; you should notice that it becomes looser. The reason we suggest that you do not weigh yourself as a mark of progress is that you may actually gain weight initially as your muscle mass increases. You may look slimmer and feel stronger, but as you gain lean muscle mass, you may have a small increase in body weight.
10. *Most importantly, enjoy yourself!* Working out is an excellent way to focus on you. It is a time where you can forget about the daily grind and think about nothing but feeling confident, healthy, and strong!



APPENDIX 2. Exercise Photo Guide

For all exercise, please remember:

- Keep your abdominals contracted during every exercise.
- Breathe correctly—exhale with exertion.
- Use a mirror to check posture.
- Maintain good posture: proper alignment of back, shoulders, and neck.
- Do not lock out your arms or legs.
- Keep your knees slightly bent.
- Do not break your form to lift a weight—simply decrease the amount of weight.
- Use slow and controlled movement during the entire repetition.
- Contact us if you experience any sharp pain in your joints or muscles and stop exercises.

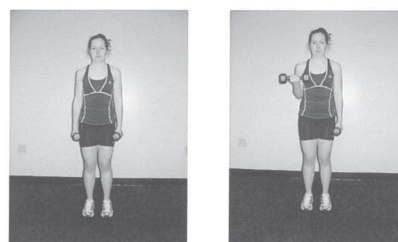
Arnold Press (Standing)

Shoulders



1. Hold weights in front with palms facing shoulders.
2. Lift dumbbells by bringing elbows outward; palms should now face forward.
3. Lift dumbbells straight up and stop before elbows are locked.

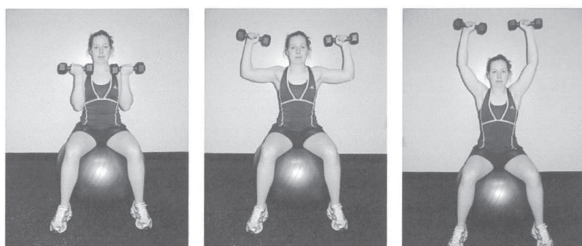
Bicep Curl (standing)



1. Have arms at your sides with palms facing inward.
2. Lift dumbbell while rotating palm to end of range.
3. Elbows should remain stationary and close to your side.
4. When returning to start position, do not lock elbow.

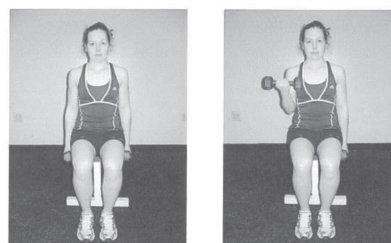
Arnold Press (seated)

Shoulders



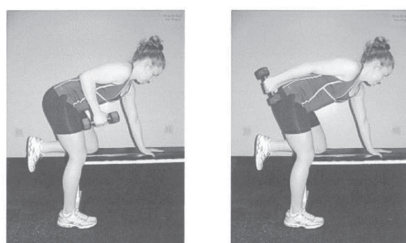
1. This is the same as the previous exercise regarding the movement of the arms and shoulders.
2. Sit on a ball and be sure your back stays straight.
3. Keep abdominals contracted so you do not roll on the ball.

Bicep Curl (seated)



1. This is the same as the previous exercise except that you will be seated on a bench or ball.
2. Keep your back straight and abdominals contracted so ball does not roll.
3. Keep elbows stationary next to the body.

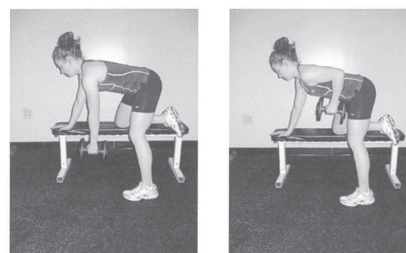
Triceps Kickback



1. Keep your elbow stationary and next to your side.
2. Lift the dumbbell until your arm makes a straight, horizontal line.
3. Be sure to keep upper arm and elbow position fixed.

Dumbbell Row

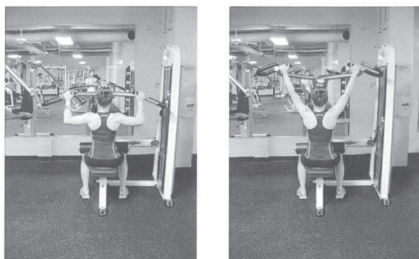
Back



1. Begin with arm hanging straight down and palm facing bench.
2. Lift dumbbell straight up until elbow is parallel with back.
3. Be sure to keep abdominal muscles contracted and avoid arching of back.

Lat Pulldown Machine

Back



1. Adjust seat height, leg cushion, and weight for your settings.
2. Grip the bar with hands placed symmetrically.
3. Pull down the bar while keeping abdominals contracted and back straight.

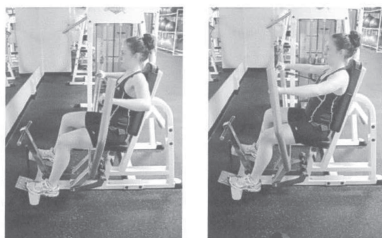
Stability Ball Squat

Legs and Buttocks



1. Start with the center of the ball on the small of your back and knees bent slightly.
2. Squat down by bending your knees and letting the ball roll up your back.
3. You should feel this mainly in your buttocks, thighs, and hamstrings.

Chest Press Machine



1. Adjust seat height and weight for your settings.
2. Sit down, push lower bar forward with foot, and gently ease weight onto upper body with hand grips.
3. Press handles forward to straighten arm without locking elbows.
4. Return with hands until gentle stretch is felt on chest.

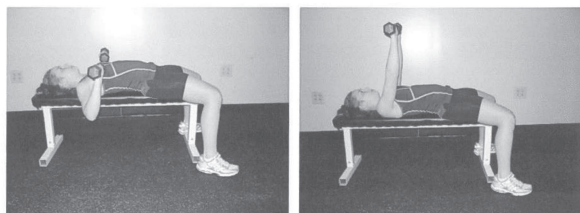
Stability Ball Squat (deep)

Legs and Buttocks



1. This is the same as the previous exercise except now you will place your feet farther from the wall making the movement more difficult.
2. Next, you will squat down until your thighs are parallel to the ground and your knees are directly above your feet.
3. Be sure to keep abdominals contracted to protect back.

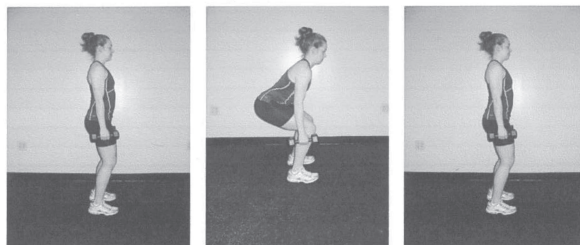
Dumbbell Chest Press



1. Lay flat on bench with arms facing straight up and palms facing feet. You should feel a gentle stretch across your chest.
2. Press dumbbells toward the ceiling until elbows are almost locked.
3. BE CAREFUL with how you set your weights down after the last rep of this exercise to avoid injury to shoulders or back.

Dumbbell Regular Squat

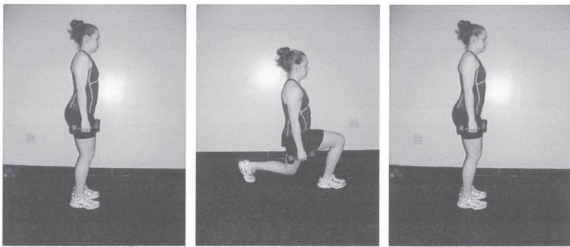
Legs and Buttocks



1. Start with feet together and proper posture.
2. Squat down by bending knees and leaning forward.
3. Keep your knees from going in front of your toes and abdominals contracted.
4. Maintain a straight back and try not to look down.

Lunge With Dumbbells

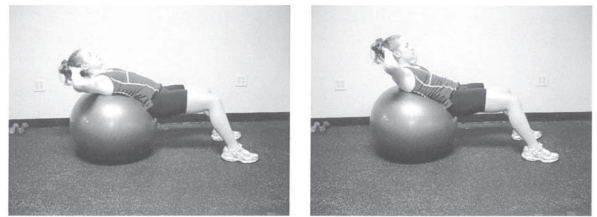
Legs and Buttocks



1. Start with feet together and proper posture.
2. Take a fairly big step forward and lower your body straight down until your hind knee is almost to the floor and your front knee is above your ankle joint.
3. Return to the start by pushing off with front heel; It is VERY IMPORTANT to not lean forward or go over your toes with your knee!

Stability Ball Crunch

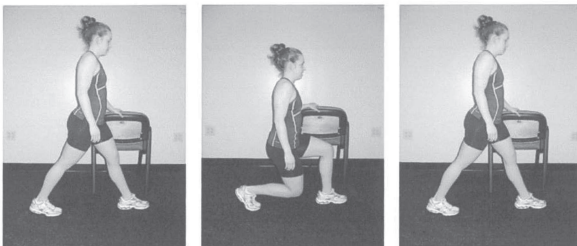
Abdominals



1. Lay with the middle of your back on the stability ball
2. Similar to the last exercise, touch temples with fingers or cross arms over chest
3. Lift shoulders upward by contracting abdominals
4. Again, do not pull your head up with your hands and stress neck

Lunge With Chair

Legs and Buttocks



1. This is the same as the previous exercise except that now you will not have dumbbells and you can use a chair or counter/dresser to help keep your balance.
2. Remember not to lean forward and keep the front knee directly above the front foot without going past the toes.

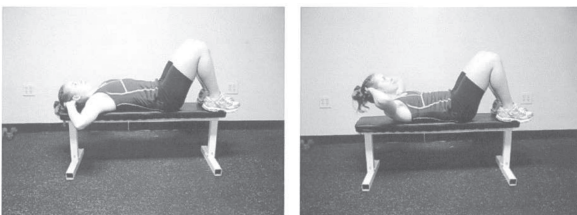
Abdominals/Obliques



1. Find some rubber tubing with a handle at the end
2. Attach it to something stationary
3. Hold the handle with both hands in front of your upper abs and move your hands away from body to straighten arms
4. Remember to keep abdominals contracted and proper posture

Crunch

Abdominals



1. Lay on a bench or yoga mat with your knees up and fingers touching your temples or arms crossed in front of chest.
2. Lift shoulders off surface by contracting abdominals.
3. Do not pull your head up with your hands and stress the neck.